Dr. Bharadwaj Chowdary Mummaneni

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Professional Summary

Quantum computing researcher with 3+ years of experience developing quantum algorithms and working with quantum hardware platforms. Currently leading a research group of 7 researchers focused on quantum simulations, where I shape the research direction and manage collaborations with industry partners and academic institutions. My work focuses on translating quantum algorithms from theoretical concepts to practical hardware implementations. I have experience communicating quantum computing concepts to diverse audiences, including researchers, industry partners, and academic groups. Published work spans variational algorithms, quantum signal processing, and neural quantum states, with practical experience deploying and benchmarking algorithms on IBM quantum computers and D-Wave systems.

Technical Skills

Quantum Computing: Variational Quantum Algorithms, Subspace Diagonalization Methods, Block Encoding, Quantum Signal Processing, Quantum Error Mitigation

Programming & Frameworks: Python, Qiskit, PennyLane, PyTorch, Linux

Quantum Hardware: IBM Quantum Systems, D-Wave Quantum Annealing Systems, IONQ

Classical Methods: Tensor Networks, Neural Quantum States, Machine Learning, Genetic Algorithms, Ab Initio Software

Professional Skills: Project Management, Technical Leadership, Workshops & Teaching, Cross-functional Collaboration, Technical Writing

Professional Experience

Quantum Algorithm Researcher | Fraunhofer IAO, Stuttgart, Germany

August 2022 – Present

Research Leadership & Collaboration

- Lead state-funded (SEQUOIA-II, KQCBW) quantum computing projects, coordinating collaborations with industry partners including Simerics, Dieter Schwarz Stiftung, and EnBW, as well as researchers at University of Stuttgart
- Worked directly with partners to identify potential quantum computing applications, translating technical requirements into research objectives and guiding projects from conceptual development through hardware validation
- Deliver technical workshops, demonstrations, and presentations on quantum computing capabilities and limitations, adapting content to audience technical background and project requirements

Quantum Algorithm Development:

- Developed quantum optimization algorithms for industrial applications including vehicle charging optimization, truck routing problems (LiDL), and optimal resource allocation of resources (EnBW) and implementing annealing, QAOA and other variational quantum algorithm approaches
- Designed and benchmarked quantum annealing based feature selection algorithms using mutual information metrics (second and third order), with experimental validation on D-Wave hardware platforms
- Applied subspace diagonalization methods and sample-based Krylov approaches for molecular ground state energy calculations, demonstrating significant performance improvements over standard variational algorithms for photovoltaic material systems

Advanced Quantum Methods:

- Co-developed variational Block-Encoding protocols combined with quantum signal processing, providing improved resource efficiency compared to traditional Linear Combination of Unitaries methods
- Investigated transformer architectures and restricted Boltzmann machines as neural quantum state representations for ground and excited state energy calculations
- Developed interpretability methods for neural quantum states in chemical spin systems, providing insights into how these models encode quantum information

Teaching & Knowledge Sharing:

- Developed and deliver quantum computing workshops and training programs for partner organizations, creating educational materials that support independent capability development
- Deployed and benchmarked quantum algorithms on IBM superconducting quantum computers and D-Wave annealing systems, addressing practical implementation challenges in transitioning research code to hardware platforms

Education

PhD in Condensed Matter Physics | Technische Universität Kaiserslautern, Germany

June 2018 – June 2022

- Thesis: Ultrafast Spin Dynamics in Molecular Magnets
- Developed theoretical methodology for quantum logic operations in molecular magnets, collaborating with international experimental groups to validate theoretical predictions
- Collaborated with experimental researchers to provide theoretical frameworks for understanding observed physical properties in magnetic systems

Master of Science in Physics | Sri Sathya Sai Institute of Higher Learning, India

June 2013 – April 2018

- Focus: Material Science and Density Functional Theory studies
- 5 conference publications on thermoelectric materials and ultrafast spectroscopy, investigating approaches to improve material efficiency

Selected Publications

Quantum Computing & Optimization:

- Rullkötter, L., **Mummaneni, B.**, et al. "Resource-Efficient Variational Block-Encoding"

- **Mummaneni, B.** "Hidden Unit Interpretability in RBM Quantum States: Encoding Antiferromagnetic Order in Heisenberg Spin Rings" (Under review)

- Sturm, A., **Mummaneni, B.**, Rullkötter, L. "Unlocking Quantum Optimization: A Use Case Study on NISQ Systems"

- Pranjic, D., **Mummaneni, B.C.**, et al. "Quantum Annealing based Feature Selection in Machine Learning"

- Ram, S.S., **Mummaneni, B.**, et al. "Multi-Qubit Superconducting Architectures: A Comprehensive Study of Design and Analysis of Transmon and Fluxonium Qubits"

- **Mummaneni, B.C.**, et al. "Laser controlled implementation of CNOT, Hadamard, SWAP, and Pauli gates in a 3d-4f molecular magnet." *J. Phys. Chem. Lett.*